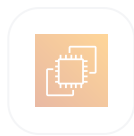


LionX compute: Amazon ECS

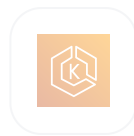
ADR-0001. The compute platform decision behind the AWS migration.



ECS Fargate



ECS on EC2



EKS

proposed

2026-06-26 Sofiane DJERBI, Bayarbileg BATBILEG, Rui SOUSA CALDAS, Laurent QUILLÉROU

The decision, and what drives it

One choice to make: where LionX compute runs on AWS.

THE WORKLOAD



8 services

long-running, plus cron



8 vCPU, 32 GB

per environment



5 environments

integration to production

Networking, CI/CD, observability and IAM all hang off this choice.

WHAT DRIVES THE DECISION

1

End-2026 cutover.

The platform has to fit the migration runway, not the other way around.

2

A rewrite either way.

The deployment layer is rewritten regardless. Today's cluster is a delivery vehicle, not a commitment.

3

Audit surface stays minimal.

Every extra component is more to audit, scan, log and upgrade.

4

A single workload.

No co-tenants to amortize a cluster's overhead against.

Three options

Same workload on each, judged on what costs engineer time.

	 ECS Fargate ✓	 ECS on EC2	 EKS standard to Auto Mode
KUBERNETES TO RUN	● none	● none	● cluster, upgrades, GitOps
NODE OPERATIONS	● AWS's job	● ours: AMIs, patching, CVEs	● ours, or AWS's with Auto Mode
AUDIT SURFACE	● CloudTrail only	● plus OS-level evidence	● plus Kubernetes audit logs
ECOSYSTEM	● cron via EventBridge, no Helm	● cron via EventBridge, no Helm	● CronJob, Helm, operators
FIT FOR LIONX	● runs it all, nothing unused	● runs everything, ops attached	● richer ecosystem, unused

The cheapest bill buys the most ops. Numbers on the next slide, full pros and cons in the ADR.

The numbers

Yearly, for the LionX footprint. Zurich list prices with a 1-year Savings Plan, ops at a conservative \$1,000 a day.

	 ECS Fargate ✓	 ECS on EC2	 EKS standard to Auto Mode
INFRA, PER YEAR	~\$4.6K	~\$3.1K	~\$4.3K to 5.2K
OPS, HOURS PER MONTH	~2 to 4	~10 to 20	~15 to 60
OPS, PER YEAR	~\$3K to 6K	~\$15K to 30K	~\$23K to 90K
FULLY LOADED, PER YEAR	● ~\$8K to 11K	● ~\$18K to 33K	● ~\$27K to 95K

Fully loaded, Fargate is the cheapest option, not the priciest. Ops hours decide it.

eu-central-2 list prices, AWS Pricing API, July 2026. Ops hours are assumptions from industry write-ups, detailed per EKS variant in the ADR.

Decision: ECS Fargate

The smallest operational surface that runs all of LionX.



ECS Fargate ✓

all eight services and the scheduled tasks, one runtime, zero nodes

Zero Kubernetes to run

no cluster, no CNI, no GitOps reconciler,
no upgrade calendar

One audit surface

CloudTrail end to end, host patching is
AWS's responsibility

Nothing blocks landing

no cluster standup on the critical path to
end 2026

Accepted trade-offs. Each schedule is an EventBridge rule plus an IAM role, more Terraform than a CronJob. The ecosystem is narrower than Kubernetes, and the difference would sit unused.

Next steps

What we ask of you today.



Questions and objections

here and now, let's settle them in the room



Ratify today

the ADR moves from proposed to accepted



PoC

pipeline, IAM boundaries and the cron path, end to end in the new AWS account

The ask today: your GO on ECS Fargate.

Any questions? Let's settle them now and call it.